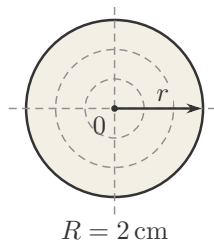


内部导热

$$\rho(C)c_p(C)\frac{\partial\theta}{\partial t} = \frac{1}{r}\frac{\partial}{\partial r}\left[rk(C)\frac{\partial\theta}{\partial r}\right]$$

$C \rightarrow k, \rho c_p$ 含水率改变热响应

(a) 中部径向截面

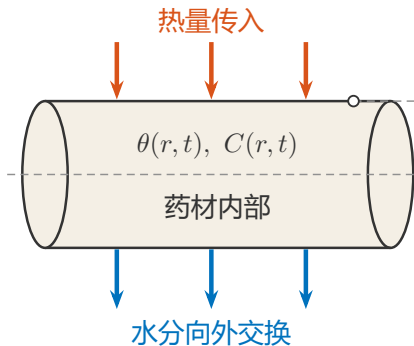


$\theta(r, t), C(r, t)$

中心对称

$$\partial_r \theta(0, t) = \partial_r C(0, t) = 0$$

(b) 侧表面传热传质



水分向外交换

固定半径, 轴对称径向近似

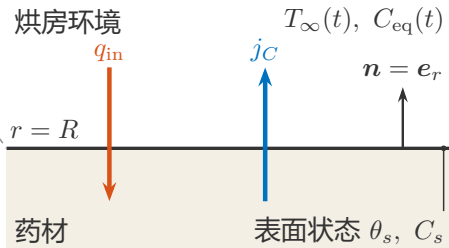
$L = 25 \text{ cm}$

内部水分扩散

$$\frac{\partial C}{\partial t} = \frac{1}{r}\frac{\partial}{\partial r}\left[rD(C, \theta)\frac{\partial C}{\partial r}\right]$$

$(C, \theta) \rightarrow D$ 温湿状态改变扩散

(c) 表面局部交换



$$q_{in} = h(T_{\infty} - \theta_s)$$

$$j_C = h_m(C_s - C_{eq})$$

$$C_{eq}(t) := Y_{\infty}(t)$$